Design Document

Streetify

Web-Based Transportation System

User Account Management (IT25102208 Nayanamina A.R.L.P.)	Ride Booking and Management (IT25102207 Dharmakeerthi W.A.C.B.)
Driver Trip Request Management (IT25102241 Senaka K.A.T.)	Payment and Receipt Management (IT25102225 Edirisinghe E.A.R.N.D.)
Review, Rating, and Feedback Management (IT25102193 Weerasingha W.A.M.B.)	Administration and System Control (IT25102240 Rammandalagedara R.V.S.)

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Table of Contents

3 Agile Sprint Summaries

4 Final Use Case Diagram

5 Class Diagram

6 - 9 Activity Diagrams

10 - 11 Ethical Considerations

12 Simple Timeline

13 Conclusion

Agile Sprint Summaries

1.1 Summary of Completed Sprints (Weeks 3 – 9)

- Sprint 1: Foundation, Architecture & Database (Weeks 3-4)
  - Deliverables: Finalized EER Diagram, Relational Schema Mapping, and MSSQL Database Creation.
  - Activities: Set up the StreetifyDB in MSSQL. Configured Spring Boot environments and mapped entities (User, Trip, Vehicle).
  - Task Allocation: R.V.S. Rammandalagedara & A.R.L.P. Nayanamina configured database schemas; W.A.C.B. Dharmakeerthi validated EER models against Spring Data JPA requirements.

- Sprint 2: Core Backend Services & Authentication (Weeks 5-7)
  - Deliverables: JWT Security middleware, User/Driver Registration, and Document Upload modules.
  - Activities: Implemented strictly layered N-Tier architecture (Controllers, Services, DAOs). Configured Spring Security and Role-Based Access Control (RBAC).
  - Task Allocation: A.R.L.P. Nayanamina developed Auth endpoints; K.A.T. Senaka built initial Trip Progress entity tracking.

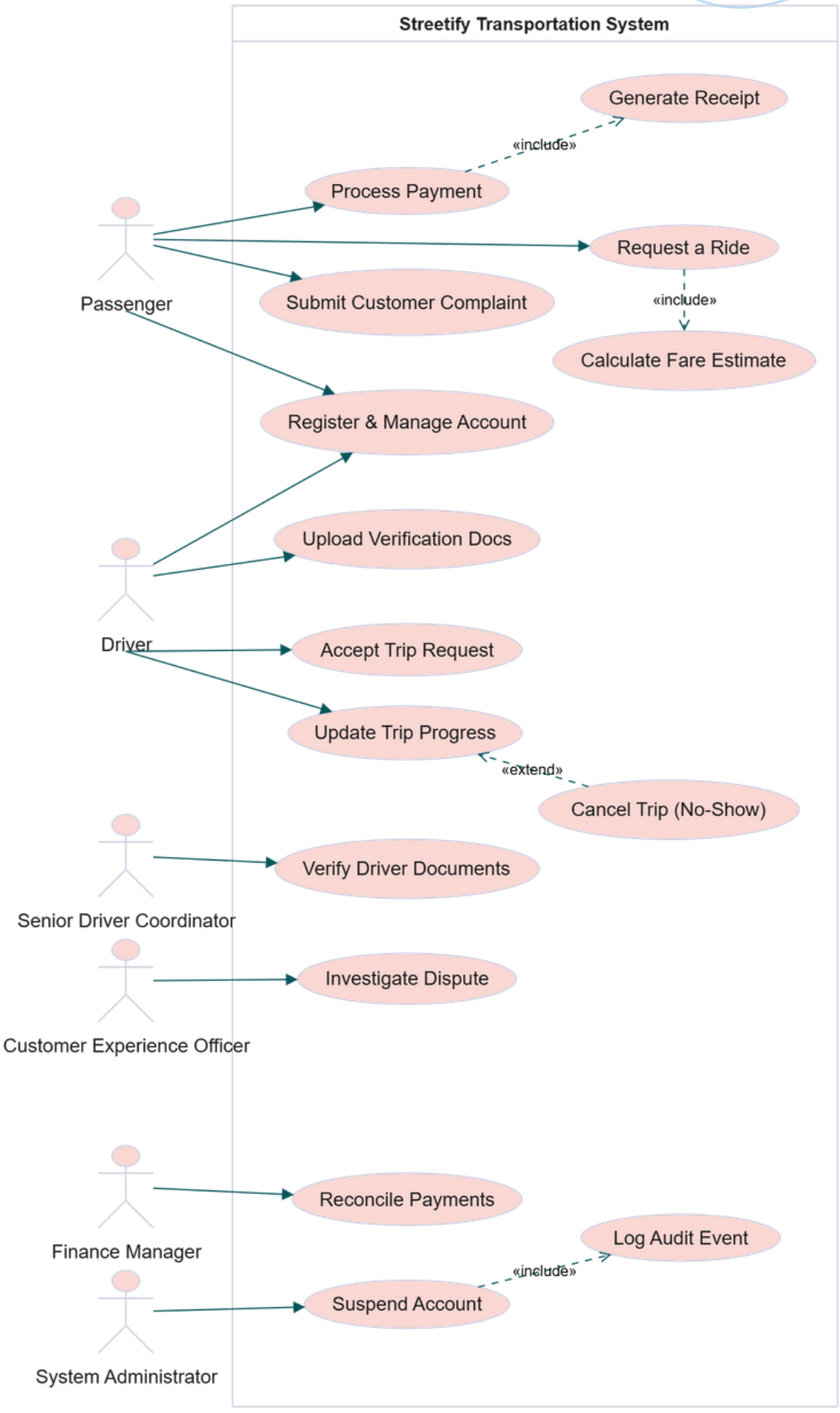
- Sprint 3: Real-Time Engine & Booking Flows (Weeks 8-9)
  - Deliverables: WebSockets integration, Live Dispatching, and Payment Gateway mockups.
  - Activities: Implemented @EnableWebSocketMessageBroker for live GPS tracking. Created the Trip state machine (Requested $\rightarrow$ Assigned $\rightarrow$ Completed).
  - Task Allocation: W.A.C.B. Dharmakeerthi completed the Ride Booking algorithms; E.A.R.N.D. Edirisinghe integrated the Payment module; W.A.M.B. Weerasingha designed the support UI templates.

1.2 Summary of Sprints to be Completed (Backlog) (Weeks 10 – 14)

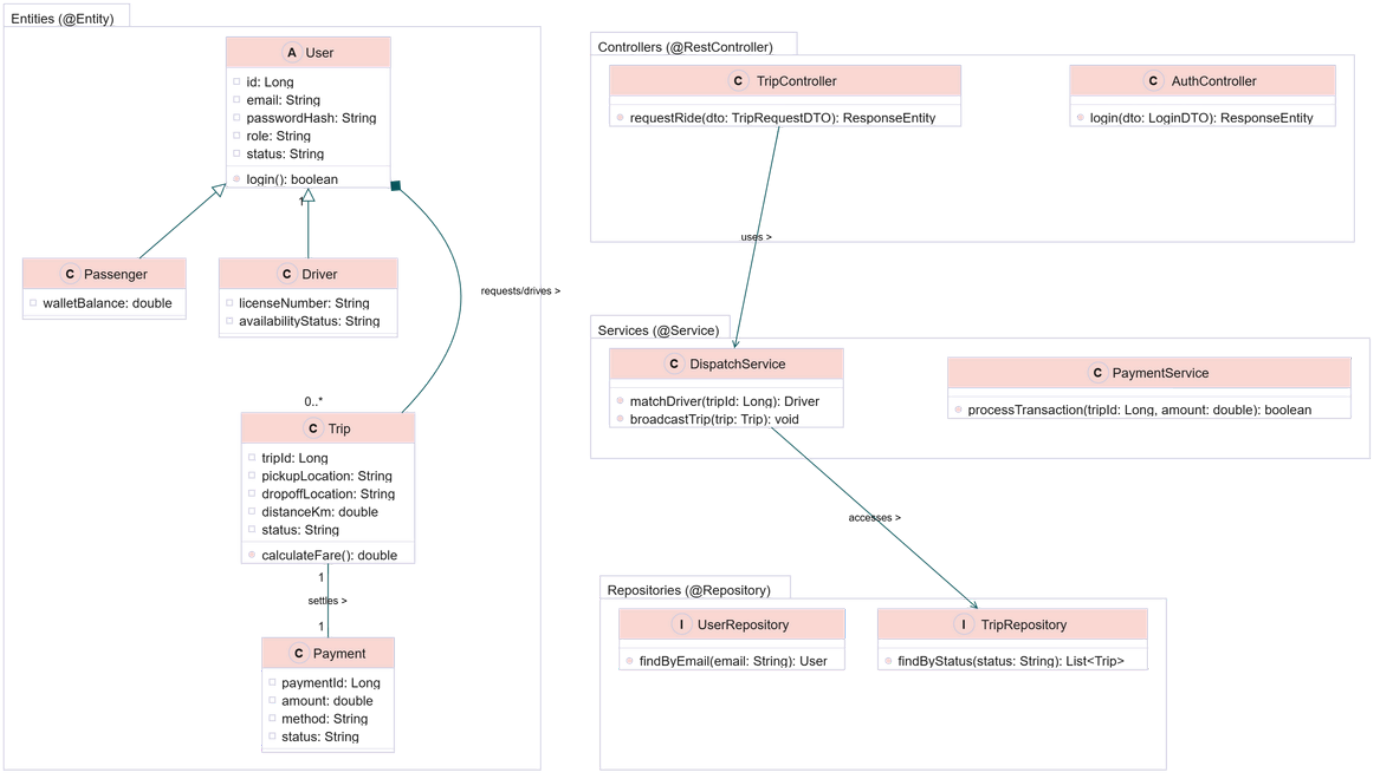
- Sprint 4: Integration, Admin Governance & Refinement (Weeks 10-11)
  - Deliverables: Full integration of the Admin Dashboard and Dispute Resolution system.
  - Activities: Connect the Customer Support UI to backend APIs. Finalize the Audit Log and JWT revocation system for suspended accounts.
  - Task Allocation: R.V.S. Rammandalagedara and W.A.M.B. Weerasingha to lead integration.

- Sprint 5: Testing, Polish & Final Release (Weeks 12-13)
  - Deliverables: JUnit 5 test coverage, UI/UX responsiveness validation, Final Presentation/Viva Prep.
  - Activities: End-to-end user acceptance testing (UAT).
  - Task Allocation: All team members execute testing on their respective major modules.

Final Use Case Diagram

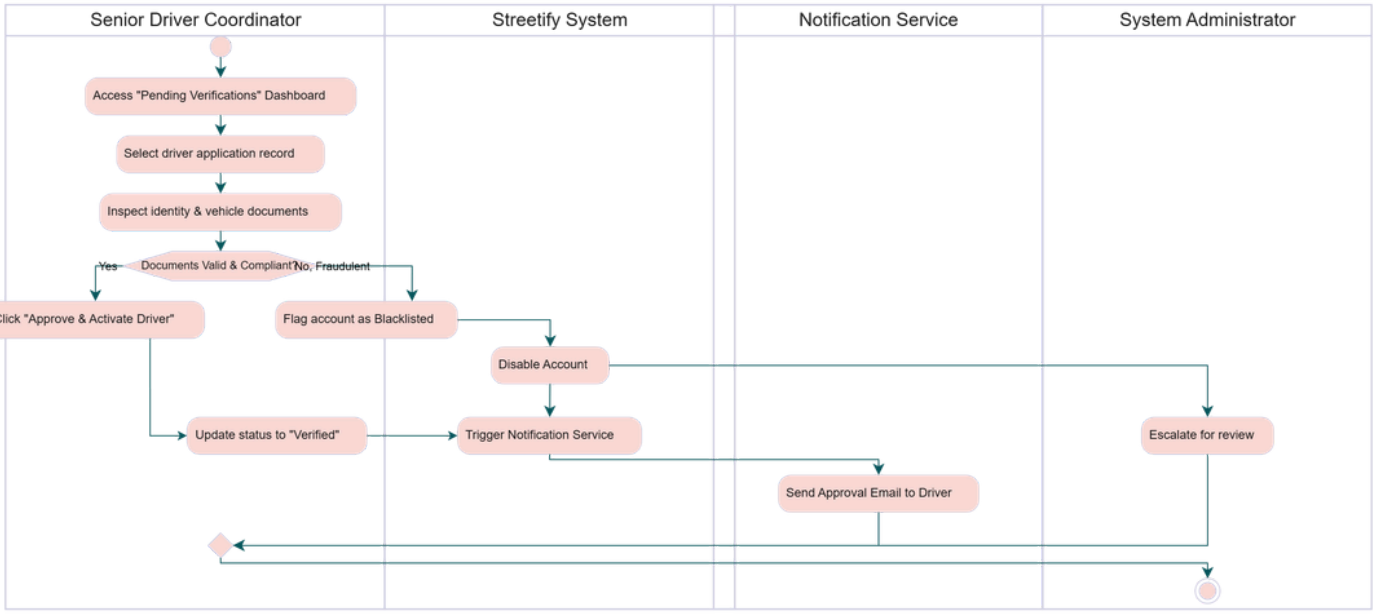


Class Diagram

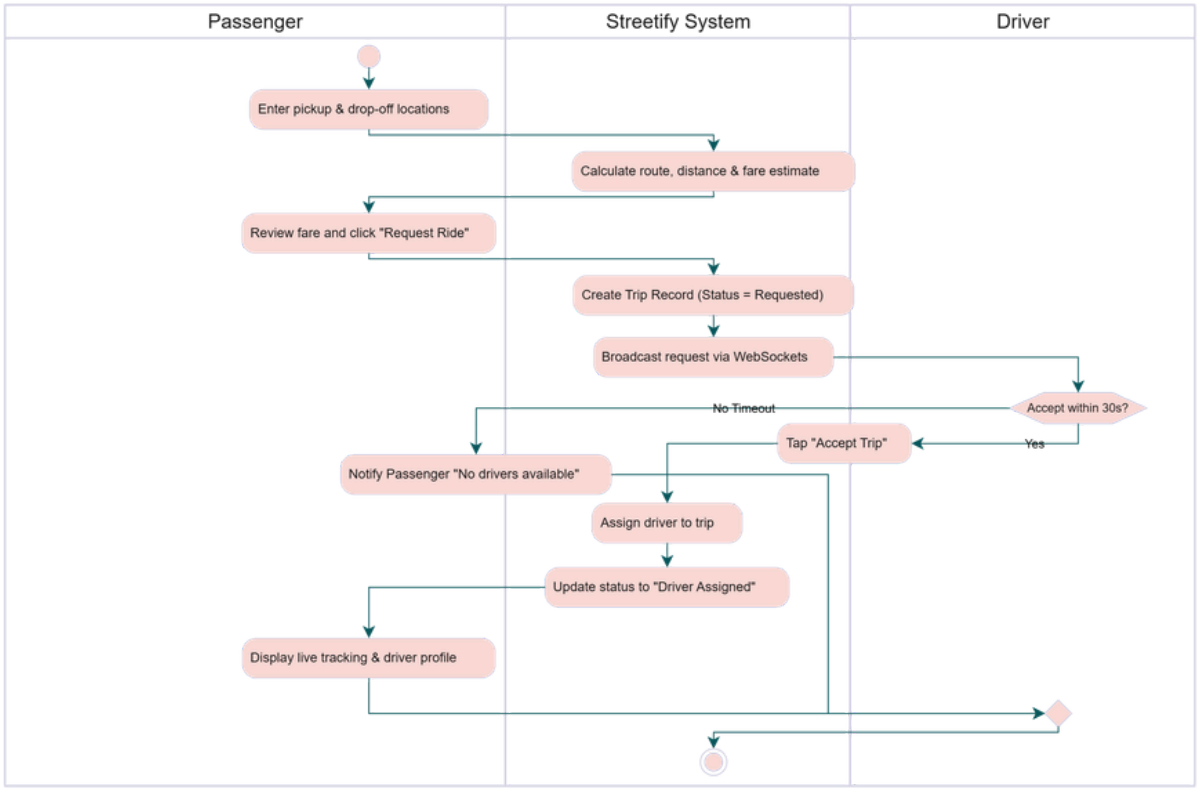


Activity Diagrams

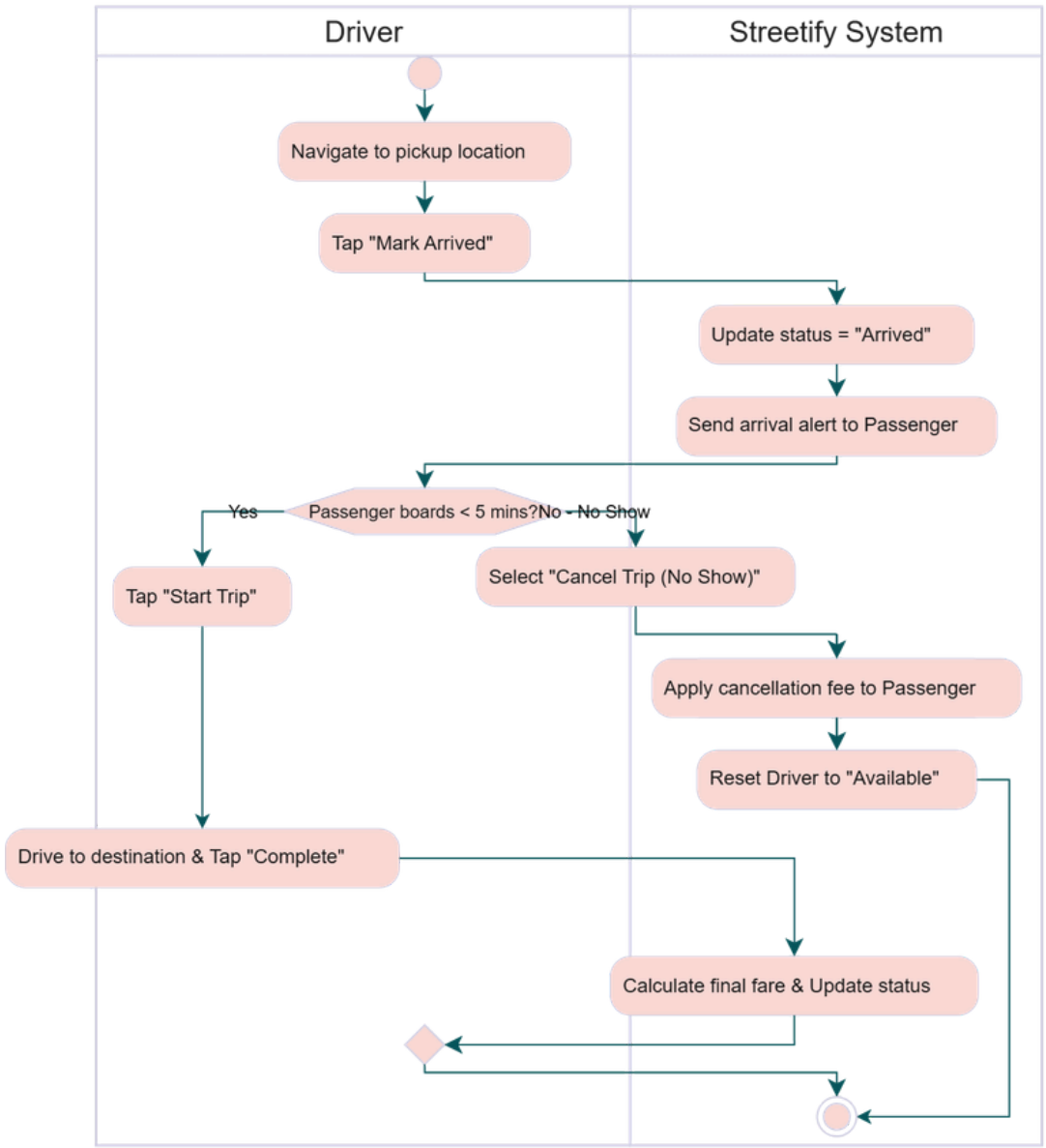
Function 1: Verify Driver Registration Documents (Nayanamina)



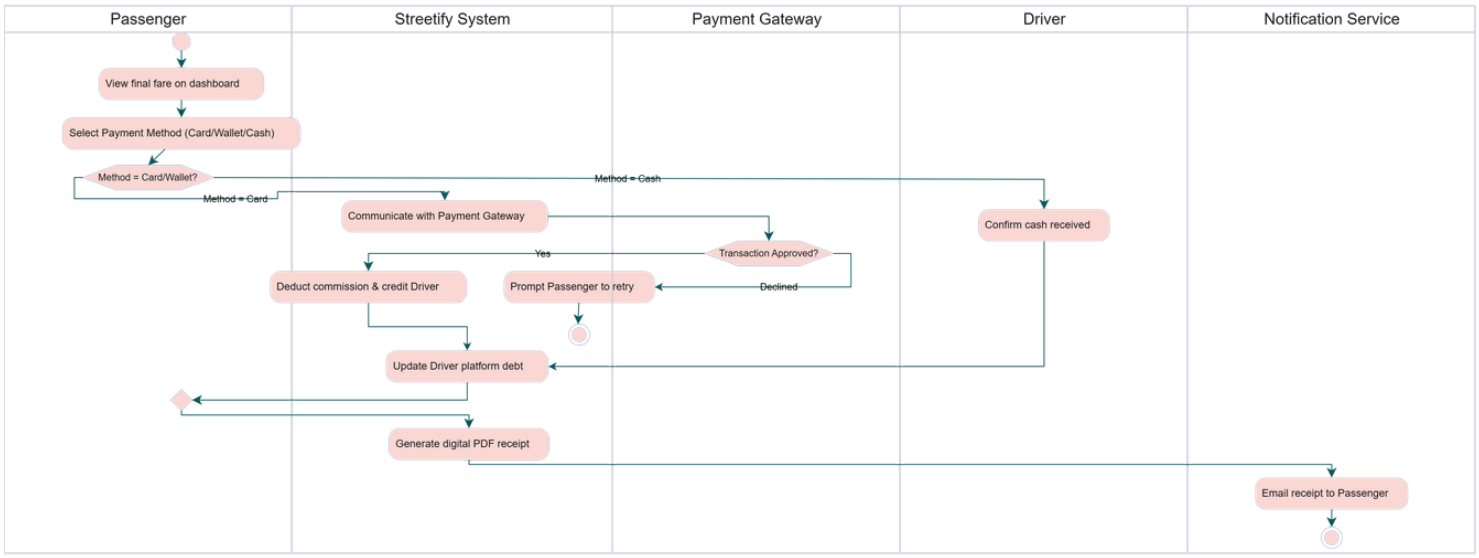
Function 2: Request a Ride and Match Driver (Dharmakeerthi)



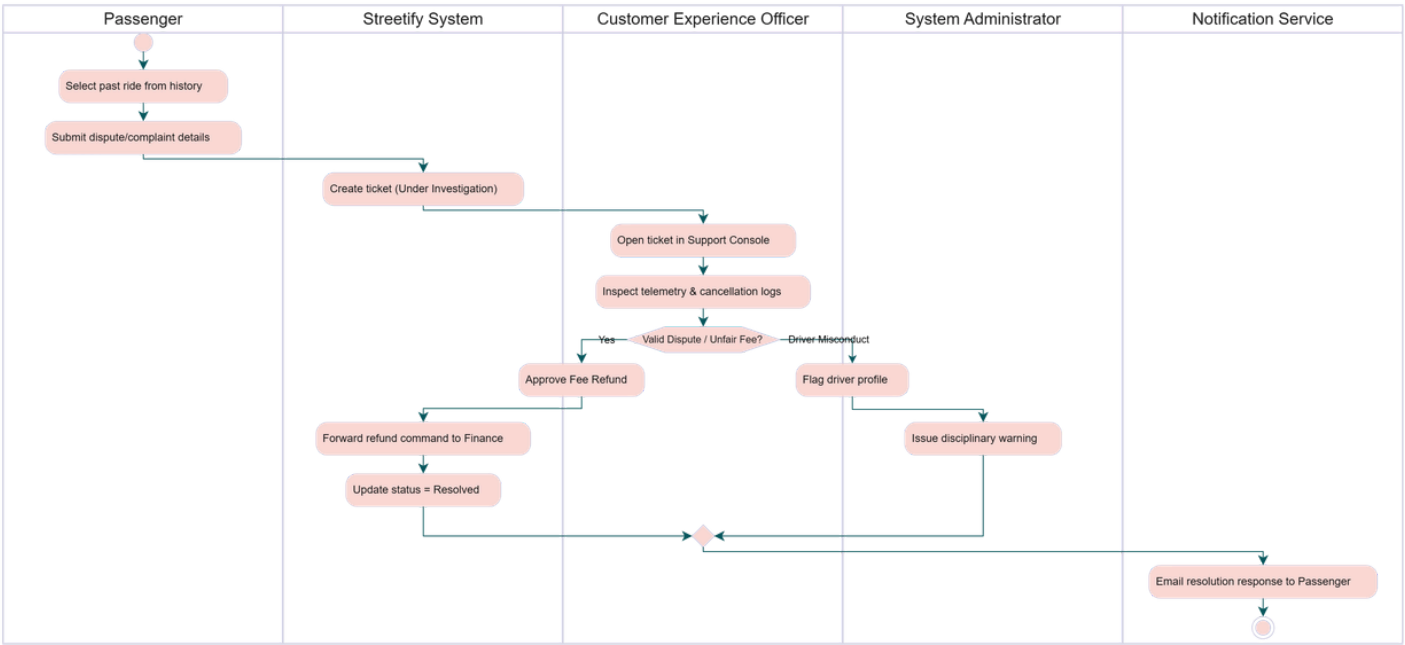
Function 3: Update Trip Progress Status (Senaka)



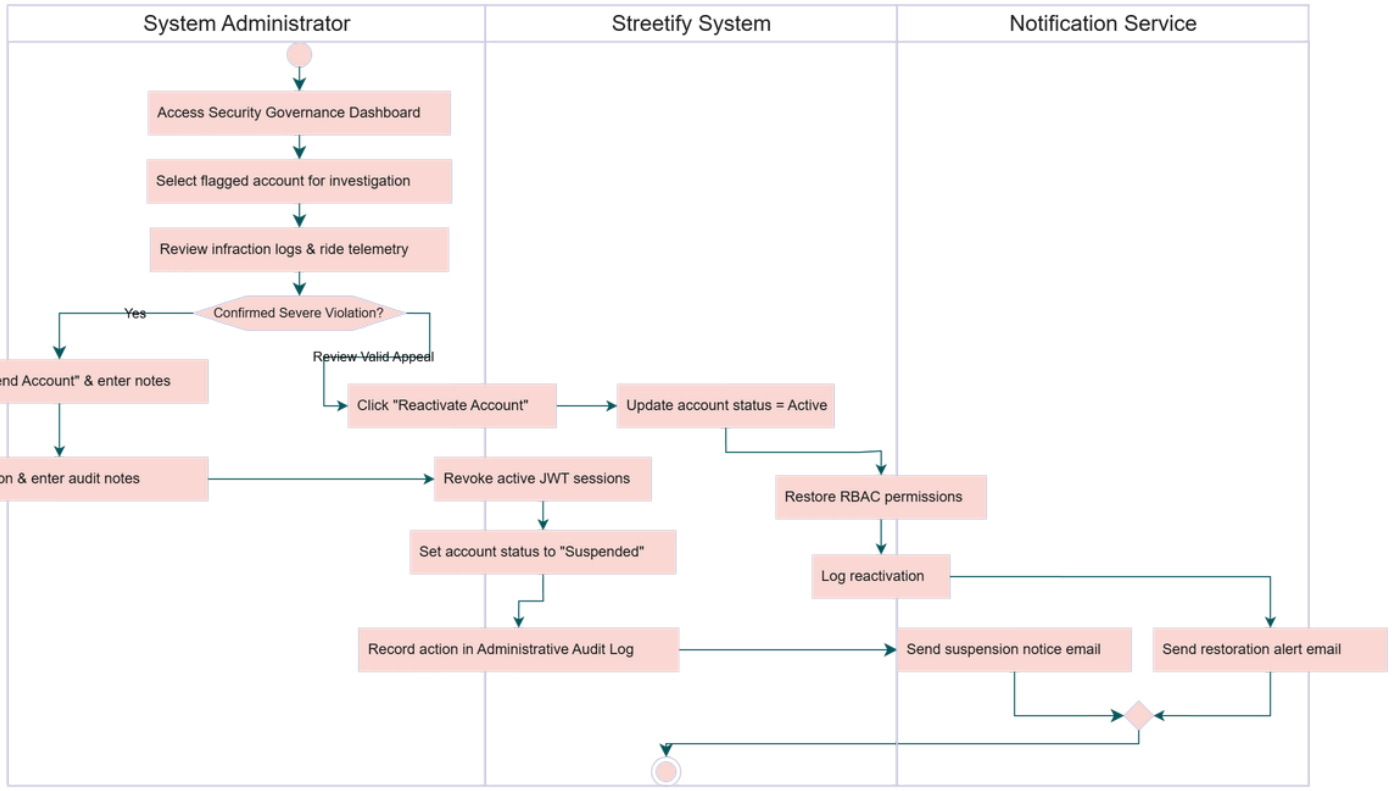
Function 4: Process Payment & Generate Receipt (Edirisinghe)



Function 5: Submit and Investigate Customer Complaint (Weerasingha)



Function 6: Suspend Account & Enforce Governance (Rammandalagedara)



Ethical Considerations

Fulfill the requirement for identifying issues and providing mitigation solutions.

5.1 Data Privacy & Confidentiality

- *Ethical Concern: Streetify processes and stores highly sensitive PII (Personally Identifiable Information), including users' home addresses, live GPS telemetry, and financial data.*
- *Mitigation Strategy: We adopted a "Privacy by Design" approach. User passwords are cryptographically hashed using bcrypt prior to database insertion. Passenger identity is dynamically masked during the ride broadcast phase (drivers only see coordinates, not passenger names). Furthermore, we utilized secure, stateless JSON Web Tokens (JWT) for authentication to prevent session hijacking.*

5.2 Security, System Integrity & Abuse Prevention

- *Ethical Concern: Malicious users could exploit the system, or administrative staff could abuse their high-level privileges to view or alter sensitive user data.*
- *Mitigation Strategy: We implemented a strict Role-Based Access Control (RBAC) architecture. To prevent internal abuse, our system includes an immutable AUDIT_LOG entity (mapped as a Weak Entity in our EER). Every time a System Administrator suspends an account or a Customer Experience Officer issues a refund, the action, timestamp, and staff ID are permanently logged. Additionally, if an account is flagged and suspended, the system immediately revokes active JWT tokens, severing live WebSocket and API access instantly to protect the platform.*

5.3 Fairness & Algorithmic Bias

- *Ethical Concern: The ride dispatching logic could inadvertently discriminate against certain passengers or neighborhoods if drivers are allowed to cherry-pick rides based on personal biases.*
- *Mitigation Strategy: The Streetify backend utilizes a strict proximity-based algorithm (calculating the Haversine distance) to broadcast rides to the nearest drivers. To enforce fairness, the driver is only presented with the pickup/drop-off route and fare estimate during the 30-second acceptance window. Demographic data (passenger name, profile picture) is strictly hidden until the driver commits to accepting the ride.*

5.4 Accessibility & Usability

- *Ethical Concern: The platform must be safely usable by drivers operating vehicles in varying lighting conditions, and accessible to passengers with visual impairments.*
- *Mitigation Strategy: Our UI/UX design prototype adheres to WCAG (Web Content Accessibility Guidelines) principles. We enforce a high-contrast color scheme (Navy Blue on Soft Ash/White). Critical driver touch targets (such as "Mark Arrived" and "Start Trip" buttons) are designed extra-large to minimize cognitive load and distraction while driving.*

Simple Timeline

Timeline	Phase / Key Deliverable	Details & Planned Milestones
Weeks 3-4	**Requirement Gathering**	Client interviews, user roles, core function identification.
Weeks 5-6	**System & Architecture Design**	Class diagrams, sequence diagrams, UI wireframes.
Weeks 7-8	**Database & Environment Setup**	Schema design, database setup, initial CRUD endpoints.
Weeks 9-10	**Core Development**	Building 6 major functions and minor utility functions.
Weeks 11	**System Testing**	Unit testing, integration testing, bug fixes.
Weeks 12	**Client Feedback & Refinement**	User acceptance testing, UI polishing.
Weeks 13	**Deployment Preparation**	Server setup, documentation finalization, report review.
Weeks 14	**Final Presentation & Submission**	Final project delivery and viva presentation.

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Conclusion

Streetify is a web-based ride-hailing system for SwiftRide Lanka (Pvt) Ltd, integrating six core modules—accounts, booking, driver requests, payments, reviews, and administration—into one platform, with defined roles for passengers, drivers, admins, finance, and support to ensure efficient, transparent urban transit.

Passenger Value	Driver Benefits	Business Control	Technical Feasibility
Transparent Fares	Fair Trip Logs	Real-Time Logs	Modular OOP Design
Instant Booking	Clear Earnings	Financial Audits	Scalable Web System

Why the Project is Worth Implementing...

- **Enhanced User Experience**: Passengers gain access to instant ride bookings, transparent pre-trip fare estimations, live ride tracking, and flexible multi-channel payment processing (card, wallet, cash) accompanied by digital receipts.
- **Empowering Transportation Service Providers**: Drivers benefit from real-time trip notifications, automated trip status updates, transparent daily earning reports, and reduced idle time between bookings.
- **Operational Oversight & Financial Governance**: System administrators and finance teams receive centralized web tools to enforce platform safety policies, monitor driver verifications, resolve cancellation disputes, and generate detailed operational and financial analytical reports.

Development Methodology

1. Object-Oriented & Modular Architecture
2. Clear Task Allocation & Individual Ownership
3. Disciplined Milestone Execution
4. Rigorous Quality Assurance & Refinement

Through this structured execution strategy, Streetify delivers a highly feasible, scalable, and commercially valuable solution for Sri Lanka's transportation ecosystem.

Thank You